

ECEN 428/722 - Post Lab Report

Lab Number: 6

Lab Title: Viterbi Decoder on FPGAs

Section Number: 602

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Objective:

The objective of this lab is to design, implement, and verify a **Viterbi Decoder** on the Zybo FPGA to decode data encoded with a convolutional encoder. The lab demonstrates how the Viterbi algorithm determines the most likely transmitted bit sequence by analyzing the received encoded stream using **trellis traversal**, **Hamming distance calculations**, and **path metric updates**. The experiment emphasizes practical realization of digital communication concepts such as convolutional coding, maximum-likelihood decoding, and finite-state-machine-based hardware implementation.

Methodology:

1. Downloaded and extracted the provided **Lab6_student_code.zip** and **lab6_base.zip** files, and opened the Vivado project base.xpr.
2. Added Verilog source files including **viterbi.v**, **bmu.v**, **pmu.v**, and the testbenches **bmu_tb.v**, **pmu_tb.v**, and **viterbi_tb.v**.
3. Verified the **Branch Metric Unit (BMU)** and **Path Metric Unit (PMU)** through individual simulations using their respective testbenches to confirm correct Hamming distance and minimum-path metric computations.
4. Integrated the modules into the **Viterbi Decoder (viterbi.v)** with parameters $r = 2$, $K = 3$, and an encoded input length of 10 bits.
5. Ran behavioral simulation using **viterbi_tb.v** to observe decoded output bits and verify trellis-based decoding accuracy.
6. Synthesized and implemented the design, ensuring successful timing closure.
7. Exported hardware with bitstream to **Vitis**, generated **BOOT.bin**, and programmed the Zybo board.
8. Executed **./lab6_viterbi_test** on the board to verify correct FPGA-based decoding.

Design:

The **Viterbi Decoder** design consists of three main logic stages:

- **Branch Metric Unit (BMU):**
Calculates the Hamming distance between received parity bits and all possible state transition outputs for each trellis column.
- **Path Metric Unit (PMU):**
Accumulates and updates the path metrics by selecting the minimum-error path entering each state, thereby tracking the most probable transitions through the trellis.
- **Traceback Unit:**
Starting from the final state with the lowest path metric, backtracks through recorded transitions to reconstruct the original bit sequence.

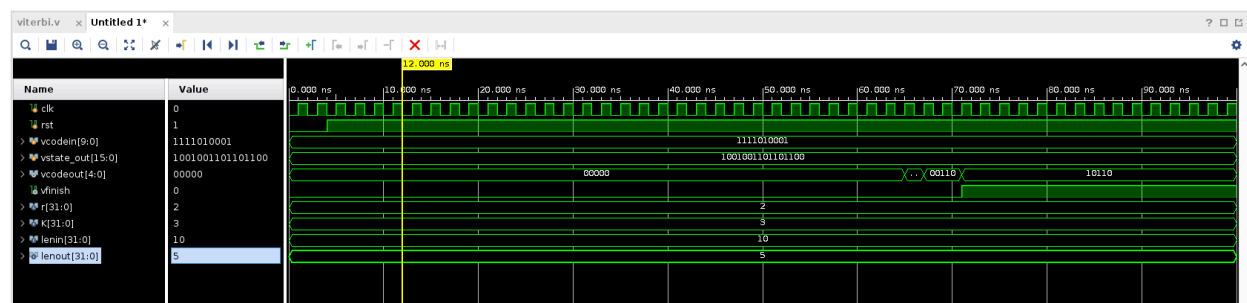
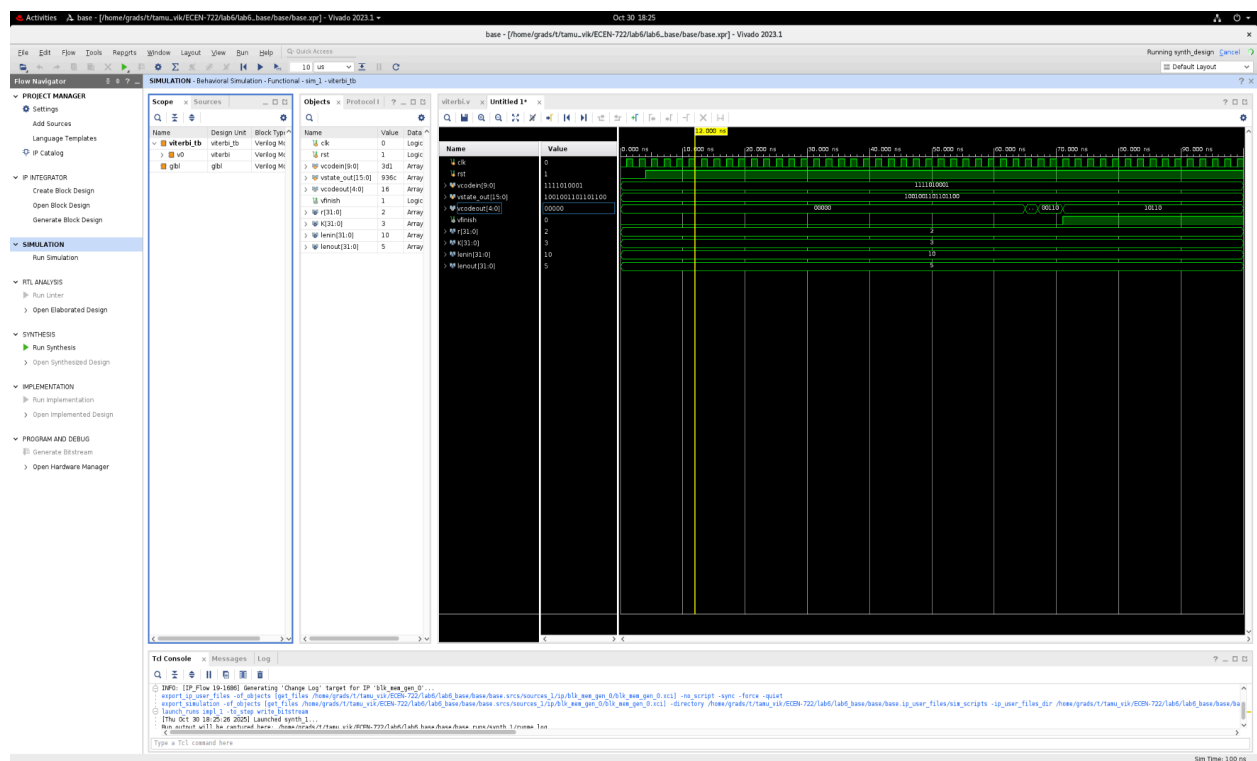
Each state transition is represented using a **finite-state machine (FSM)** corresponding to the encoder structure with $K-1 = 2$ memory elements. The decoder uses combinational and sequential logic to compute branch errors, update path metrics, and generate decoded bits in order.

Implementation:

- After verifying functional correctness through simulation, the top-level design (top_viterbi.v) was set as the top module and a True Dual-Port Block RAM (8-bit width, 65 536-depth, Read-First mode) was added via the IP catalog.
- The design was synthesized and implemented successfully in Vivado, showing proper timing, resource utilization, and power results.
- The bitstream was then exported to Vitis, where the FPGA was programmed using BOOT.bin. Running ./lab6_viterbi_test produced the decoded output sequence on the terminal, confirming that the FPGA implementation correctly recovered the transmitted data.

Results:

Behavioral simulation:

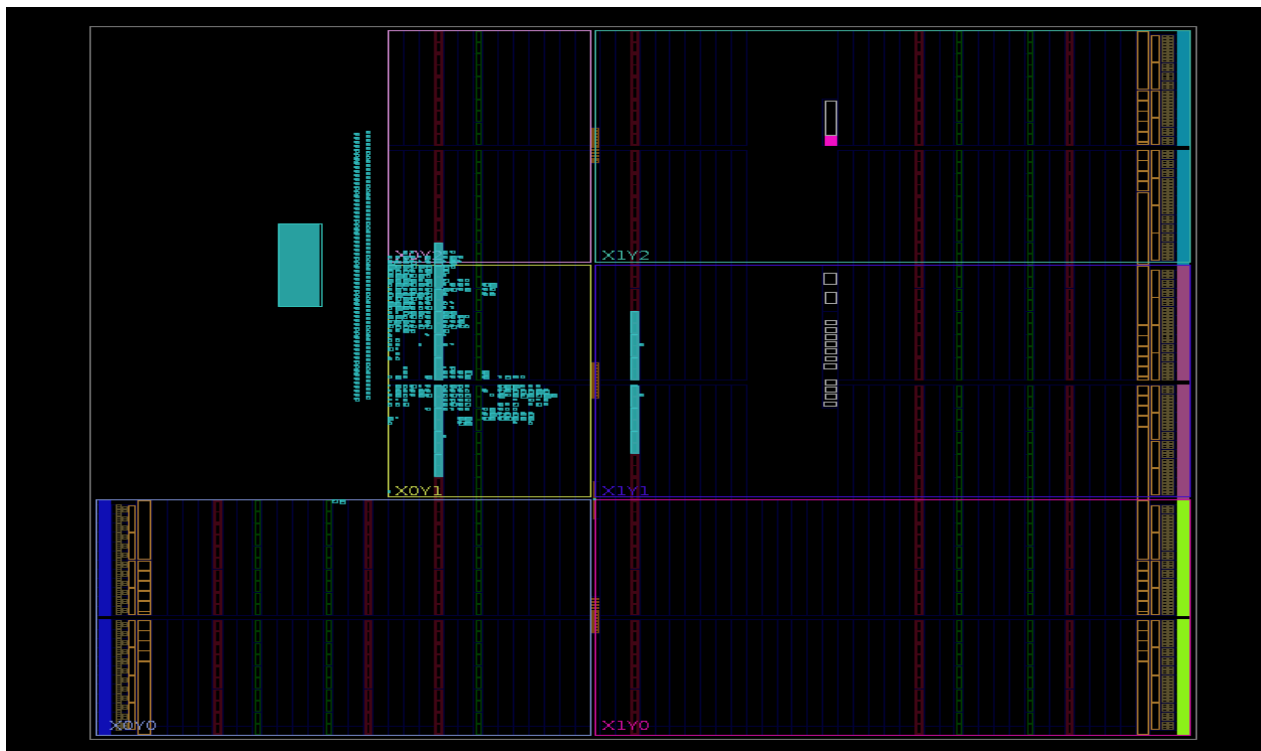


Terminal output:

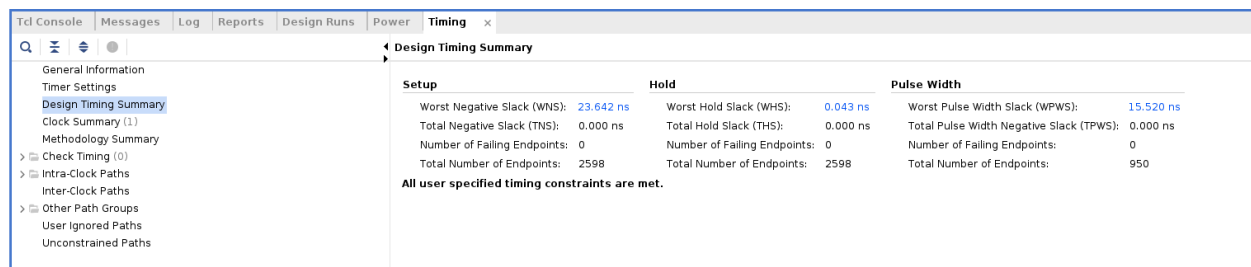
```
zynq> mount /dev/mmcblk0p1 /mnt/
FAT-fs (mmcblk0p1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.
zynq> cd /mnt/
zynq> insmod transfpfga.ko
Mapping virtual address...
The physical address is 0x43c00000
The virtual address is 0x608e0000
Registered a device with dynamic Major number of 245
Create a device file for this device with this command:
'mknod /dev/transfpfga c 245 0'.
zynq> mknod /dev/transfpfga c 245 0
zynq> ./lab6_viterbi_test
input code: 1110111101
state output: 1001011000111100
decoded code: 10011
correct code: 10011
FPGA clock cycles: 241
input code: 1110111000
state output: 0011100101101100
decoded code: 11000
correct code: 11000
FPGA clock cycles: 241
input code: 1001111100
state output: 1000110111011000
decoded code: 10110
correct code: 10110
FPGA clock cycles: 241
zynq> █
```

Additional Screenshots:

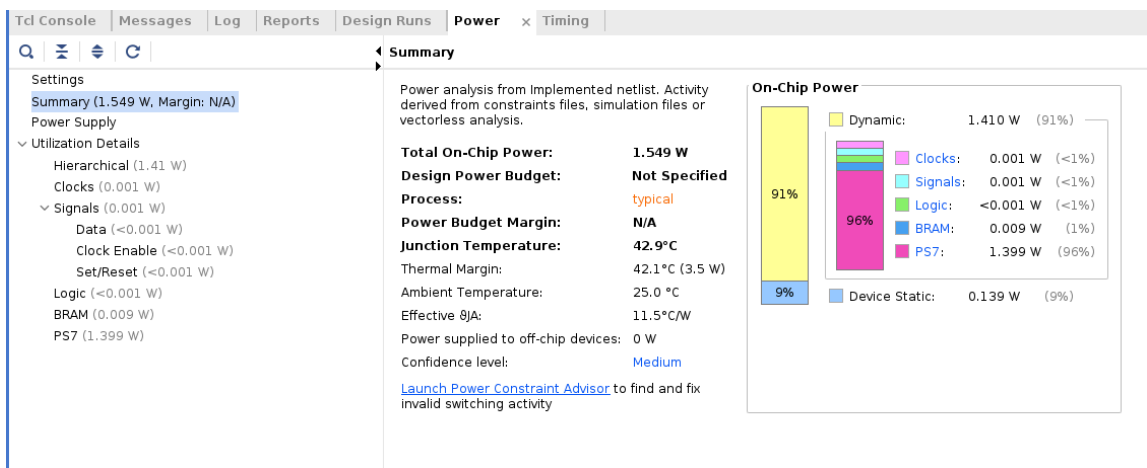
Implemented design:



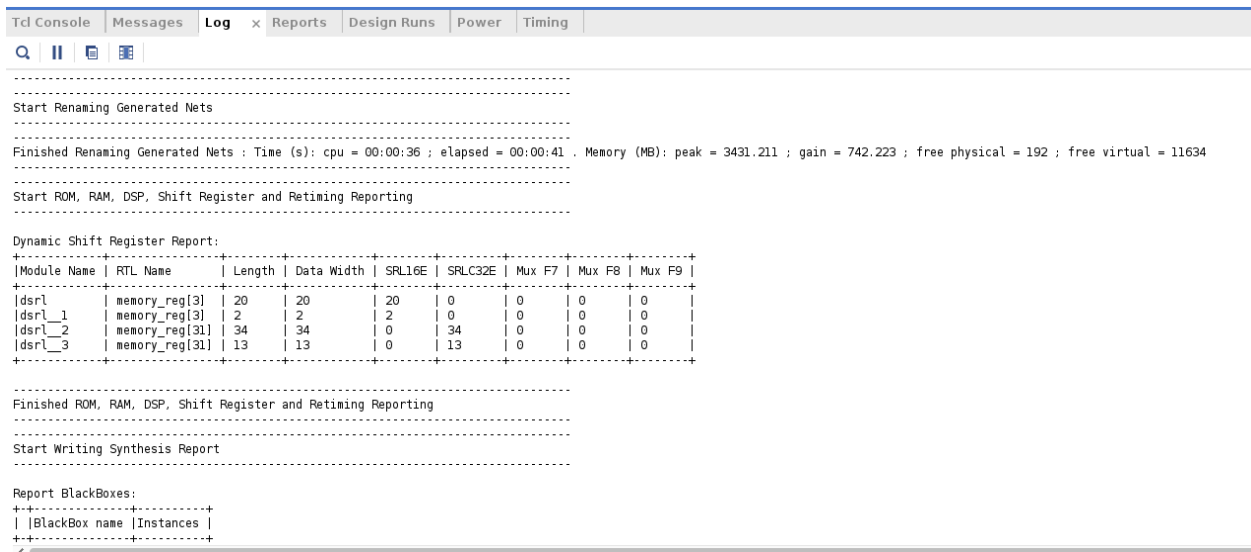
Timing summary:



Power consumption :



Hardware Utilization :



Throughput calculation:

The signal **vfinish** is asserted at **70 ns**, indicating that the decoder output becomes available at this time.

Given that the Clock Cycle Time (CCT) is 2 ns, the total number of cycles required is:

$$\text{Total number of cycles} = 70 \text{ ns} / 2 \text{ ns} = 35$$

Since the reset signal is asserted for the first two cycles, the **effective number of cycles** used for decoding is:

$$\text{Actual number of cycles} = 35 - 2 = 33$$

The **throughput** is determined using the following relation:

$$\text{Throughput} = (\text{Number of Input Bits}) / (\text{Number of Cycles} \times \text{CCT})$$

$$\text{Throughput} = 10 / (33 \times 2 \text{ ns})$$

$$\text{Throughput} = 151 \text{ Mbps}$$

Therefore, the **decoder throughput** is approximately **151 Mbps**.

Inference:

The simulation and FPGA results confirmed correct operation of the Viterbi Decoder. The BMU and PMU accurately computed branch and path metrics, while the Traceback Unit reconstructed the original bit sequence. The vfinish signal at 70 ns verified successful decoding, and the design achieved a throughput of about 151 Mbps, demonstrating efficient and reliable hardware implementation.

Conclusion:

The Viterbi Decoder was successfully designed, simulated, and implemented on the FPGA using Verilog HDL. The decoder correctly recovered the transmitted bit sequence with high accuracy and achieved a **throughput of 151 Mbps**. The lab effectively demonstrated the hardware realization of convolutional decoding and its application in digital communication systems.